# Wang Ruijie

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## Education

Aug '26 – May '31 exp. **University of Michigan, Ann Arbor**
Doctor of Philosophy in Computer Science and Engineering
Advised by Prof. Ke Wu

Aug '22 – May '26 **The Hong Kong Polytechnic University**
Bachelor of Science in Computer Science, Minor in Applied Mathematics, First Class Honours
Advised by Prof. Bo Li

Sep '19 – June '22 **Yali High School**
Track of Science, National College Entrance Examination (Gaokao)

## Research Interests

**Economics and Computation**

- Mechanism Design
- Blockchain Economics
- Fair Division

## Experiences

Nov '25 – Present **Research Collaboration**
With Dr. Aviv Yaish
- Order-sensitive auctions design

Sep '23 – Present **Undergraduate Research Assistant, PolyU COMP**
With Prof. Bo Li and Prof. Ankang Sun
- Fair allocation problem of indivisible goods and chores in graphs
- Our paper *Fair Orientations: Proportionality and Equitability* to appear at AAMAS '26

May '25 – Aug '25 **Mitacs Globalink Research Internship, Simon Fraser University**
With Prof. Yuepeng Wang
- Formal program synthesis for vector databases

Dec '23 – Jan '24 **Service Learning and Volunteer Teaching**
Da Nang, Viet Nam
- Support information technology and intercultural education at Lý Công Uẩn Primary School

May '23 – Aug '23 **Exchange Student, Technical University of Denmark**
Data Analytics, Computational Imaging and Spectroscopy

## Publications and Working Papers

Submitted **Order Auctions with Private Position Preferences**
**arXiv**
With Aviv Yaish
- We investigate two-position auctions where bidders have private preferences: "specialists" only value the first position, while "generalists" value both equally.
- Standard item-agnostic first-price auctions fail to achieve ex-post efficiency, providing only a $1/2$ expected welfare guarantee. Eliciting one extra bit of information per bidder to declare their demand is communication-optimal for efficiency. Implementing this single bit (item-specific first-price auctions) significantly improves the welfare guarantee to $1 - 1/e$.
- While VCG mechanisms achieve full efficiency, they are uniquely vulnerable to seller shill bidding, whereas first-price auctions remain robust against such manipulations.

Submitted **Settling PROPm and PROPavg in Graphical Resource Allocation**
With Ankang Sun and Bo Li
- We showed that PROPm orientations always exists and can be computed in polynomial time in multigraphs of mixed items, with the exact price of fairness of 2.
- PROPavg orientations need not exist even for simple graphs and are NP-hard to determine. Maximizing social welfare for PROPm orientations is APX-hard, and determining PO and PROPm orientations is NP-hard.

AAMAS '26 **Fair Orientations: Proportionality and Equitability**
**arXiv**, **ACM Digital Library**
With Ankang Sun and Bo Li
- We introduce proportionality, equitability, and their relaxations for the fair orientation model in graphs.
- PROP1 + fPO always exists and can be computed in polynomial time for orientations in hypergraphs, whereas PROP, PROPX, EQ, EQX, and EQ1 orientations of goods and chores and EF1 orientations of chores may not exist and are NP-hard to decide in simple graphs or multigraphs.

## Awards

March '26 **PhD First Year Award**
Computer Science and Engineering, University of Michigan, Ann Arbor

'23 – '25 **Dean's Honours List of Outstanding Students**
Faculty of Engineering, The Hong Kong Polytechnic University

Feb '25 **Mitacs Globalink Research Award**
Full fund for Mitacs Globalink Research Internship in Canada

Aug '23 **HKSAR Government Scholarship Reaching Out Award**

## Courses and Skills

Theory Design and Analysis of Algorithms, Theory of Computation, Algorithmic Economics, Cryptography, Social Computing, Game Theory, Data Structures, Graph Theory

Math Multivariable Calculus, Linear Algebra, Differential Equations, Probability and Distributions, Statistical Inference, Convex Optimization, Operations Research

Systems Operating Systems, Database, Networking, Computer Organization, Computer System Security, Kernel Programming, Compilers, Software Engineering

AI & ML Machine Learning, Artificial Intelligence, Big Data Analytics

Prog. Lang. LaTeX, Java, C/C++, Python, Lua, R, Matlab, Shell Languages, Pascal

## Personal Details

Hometown Changsha, Hunan, China

Languages English, Chinese

Sports Badminton